

Total No. of Questions : 8]

SEAT No. :

PA-1237

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[5925] 259

S.E. (Computer Engineering)

DIGITAL ELECTRONICS AND LOGIC DESIGN

(2019 Pattern) (Semester - III) (210245)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data, if necessary.

**Q1)** a) What are sequential circuits? Explain excitation table of JK flipflop. [6]

b) Convert Following Flipflops: [6]

i) SR to JK

ii) JK to D

c) What is MOD counter? Design MOD - 24 counter using 7490. [6]

OR

**Q2)** a) What are sequential circuits? Explain SR flipflop using a suitable example. [6]

b) Convert Following Flipflops: [6]

i) JK to T

ii) SR to D

c) Design sequence detector using MS JK flipflop for sequence 1101. [6]

**Q3)** a) Draw ASM chart for 2-bit UP counter using multiplexer controller method. [6]

b) Draw a block diagram of the PLA device and explain. [6]

c) Implement following Boolean function using PAL. [5]

$$F1 = \sum m(0, 2, 3, 4, 5, 6, 7, 8, 10, 11, 15)$$

$$F2 = \sum m(1, 2, 8, 12, 13)$$

OR

P.T.O.

**Q4) a)** What is an ASM Chart? Design the ASM chart for a 2-bit binary counter having one enable line E such that when: [6]

E = 1 (count enabled) and

E = 0 (counting is disabled).

**b)** Implement 3 bit binary to gray code converter using PLA. [6]

**c)** A combinational Circuit is defined by the following function: [5]

$$F1(A, B, C) = \sum m(0, 1, 3, 7)$$

$$F2(A, B, C) = \sum m(1, 2, 5, 6)$$

Implement this circuit with PLA.

**Q5) a)** Explain the operation of TTL NAND gate. [6]

**b)** Compare TTL and CMOS families and also draw CMOS-NOR Gate. [6]

**c)** Define the following terms and mention the standard values for TTL logic Family: [6]

i) Noise Margin

ii) Power Dissipation

iii) Propagation Delay

OR

**Q6) a)** Explain TTL open collector. [6]

**b)** Draw and explain the circuit diagram of the CMOS Inverter. [6]

**c)** Draw two input standard TTL NAND gate circuit and explain their operation. [6]

**Q7) a)** What is Microprocessor? Explain the system bus in brief. [6]

**b)** Which are various functional units of microprocessors? Explain ALU in brief. [6]

**c)** How Basic Arithmetic operations are performed using ALU IC 74181? [5]

OR

**Q8) a)** What is Microprocessor? Explain various operations of the microprocessor. [6]

**b)** Explain the Memory organization of the microprocessor. [6]

**c)** Explain the 4-bit Multiplier circuit using ALU and shift registers in brief. [5]

